Paper 2: Dual Impedance Networks and Metabolic Scaling
Vascular Architecture, Organ Health, Fractal Cost, and Kleiber's Law under the Minimum Reflection Principle

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We ask: given a blood vessel with impedance $Z_\mathrm{v} = 8\mu L/\pi r^4$ governed by vascular C2 update, what architecture—and what metabolic scaling laws—must emerge?

Part A derives vascular architecture from the Minimum Reflection Principle (MRP). A blood vessel is a hydraulic transmission line. At every bifurcation, impedance mismatch produces a reflection Γ_v; C1 gives $\Gamma_\mathrm{v}^2 + T_\mathrm{v} = 1$. We derive Murray's Law ($r_p^3 = \sum r_i^3$) as the exact minimum of the vascular action $A_\mathrm{v} = \sum_j [Q_j^2 Z_j + \lambda V_j]$, with 100% numerical agreement for $n = 2, 3, 4$. Vascular C2 update ($\Delta Z_\mathrm{v} = -\eta_v \Gamma_\mathrm{v} \tau_\mathrm{wall} \rho$) operates on the same principle as neural C2 but with $\eta_v \ll \eta_n$.

Part B derives the dual-network organ model. Every organ is served by two coincident networks: vascular (Γ_v) and neural (Γ_n). Organ health $H = T_\mathrm{n} \cdot T_\mathrm{v} = (1-\Gamma_\mathrm{n}^2)(1-\Gamma_\mathrm{v}^2)$ follows from multiplicative transmission. Four failure modes are classified by which network fails. The organ impedance hierarchy ranks ten organs by their resting Γ_v^2.

Part C derives metabolic scaling. A self-similar relay hierarchy with branching number n in d dimensions produces an irreducible action $A_\mathrm{cut} \propto M^{(d-1)/d}$ (for $d = 3$: $M^{2/3}$). The Murray bifurcation ratio $\beta = n^{1/d} = 2^{1/3}$ sets length and radius contraction across levels. Kleiber's Law ($B \propto M^{3/4}$) follows from capillary invariance. Dual-network packing yields a metabolic overhead $\Omega \approx 1.33$, with $\Omega_n \approx \beta$. Multi-scale verification confirms the irreducibility theorem across cardiovascular, musculoskeletal, and 29-tissue-atlas scales.

Part D demonstrates interface universality. The same $\Gamma = (Z_2 - Z_1)/(Z_2 + Z_1)$ governs seven physical domains across 14 orders of magnitude—from synapses to power grids. The Interface Isomorphism Proposition proves that Γ depends only on boundary impedance, not bulk properties.

Circularity disclosure. Internal experiments use simulation modules built from C1–C3; they demonstrate self-consistency, not empirical validity. External tests are reported in Papers 3–5 [3–5].

Part A: From Blood Vessel to Vascular Network

I. INTRODUCTION

A blood vessel is a tube carrying fluid under pressure. Its impedance—the ratio of pressure drop to volume flow—is a physical fact set by geometry and fluid properties.

Paper 1 [2] derives neural architecture and the mind from the Minimum Reflection Principle (MRP). Yet every organ is governed by a *second* impedance network: the vasculature. The pressure wave in an artery reflects at every bifurcation, stricture, or compliance mismatch, precisely as an electromagnetic wave reflects at a transmission-line discontinuity [7, 8].

This paper derives the vascular architecture from the same MRP, then combines both networks into the dual-network organ model, and finally shows that metabolic scaling laws (Murray, Kleiber) are necessary consequences of the same irreducible topology.

The derivation follows a strictly outward chain: single vessel (Sec. II) → Murray's Law (Sec. III) → vascular C2 (Sec. V) → dual-network organ model (Sec. VI) → fractal cost (Sec. IX) → Kleiber's Law (Sec. X) → interface universality (Sec. XIII).

II. VASCULAR IMPEDANCE PHYSICS

A. The blood vessel as transmission line

A cylindrical vessel of length L and radius r carrying a Newtonian fluid of viscosity μ has steady-state impedance given by the Poiseuille law:

$$Z_\mathrm{v} = \frac{\Delta P}{Q} = \frac{8\mu L}{\pi r^4}. \tag{1}$$

In the pulsatile regime, Womersley theory adds a frequency-dependent component:

$$Z_\mathrm{v}(\omega) = \frac{8\mu L}{\pi r^4} \cdot F(\alpha), \tag{2}$$

where $\alpha = r\sqrt{\omega \rho_f / \mu}$ is the Womersley number, $F(\alpha) \to 1$ for $\alpha \ll 1$ (Poiseuille limit) and $F(\alpha) \to j\alpha^2/8$ for $\alpha \gg 1$ (inertial limit) [7].

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TABLE I. Electromagnetic–vascular transmission-line analogy.

EM quantity	Vascular	Unit
Voltage V	Pressure P	Pa
Current I	Flow Q	m^3/s
Impedance Z	$\Delta P/Q$	$\mathrm{Pa\,s/m}^3$
Γ	Γ_v	dimensionless
$T = 1 - \Gamma^2$	T_v	dimensionless

B. Three physical laws in the vascular domain

C1—Energy Conservation. $\Gamma_\mathrm{v}^2 + T_\mathrm{v} = 1$ at every vascular junction.

C2—Vascular Update.

$$\Delta Z_\mathrm{v} = -\eta_v\, \Gamma_\mathrm{v}\, \tau_\mathrm{wall}\, \rho, \tag{3}$$

where τ_wall is wall shear stress (x_in), ρ is material supply (x_out), and η_v is the vascular update rate.

C3—Dimensional Relay. Blood pressure waves carry source impedance as metadata.

C. Bifurcation reflection

At a symmetric bifurcation (n identical daughters), $Z_\mathrm{v}^\mathrm{down} = Z_\mathrm{v}^d/n$:

$$\Gamma_\mathrm{v} = \frac{Z_\mathrm{v}^d/n - Z_\mathrm{v}^p}{Z_\mathrm{v}^d/n + Z_\mathrm{v}^p}. \tag{4}$$

Pure impedance matching ($\Gamma_\mathrm{v} = 0$) gives $r_d = r_p/n^{1/4}$, which *disagrees* with Murray's Law. The resolution: MRP minimises the *total vascular action* including metabolic maintenance cost.

III. MURRAY'S LAW FROM THE VASCULAR ACTION PRINCIPLE

A. Vascular action functional

$$A_\mathrm{v} = \sum_j \left[Q_j^2\, Z_j + \lambda\, V_j\right], \tag{5}$$

where Q_j is flow, $Z_j = 8\mu L_j/\pi r_j^4$ is impedance, $V_j = \pi r_j^2 L_j$ is volume, and λ is the metabolic cost per unit volume.

B. Derivation for symmetric n-branching

Daughter contribution:

$$A_\mathrm{v}^d = \frac{Q^2\, 8\mu L_d}{n\,\pi\, r_d^4} + n\lambda\pi r_d^2 L_d. \tag{6}$$

Minimising $\partial A_\mathrm{v}^d/\partial r_d = 0$:

$$r_d^6 = \frac{16\, Q^2\, \mu}{n^2\, \lambda\, \pi^2}, \qquad r_p^6 = \frac{16\, Q^2\, \mu}{\lambda\, \pi^2}. \tag{7}$$

Taking the ratio:

Theorem 1 (Murray's Law from MRP)**.** *For symmetric n-branching of a vessel with parent radius r_p, minimisation of the vascular action $A_v = \sum_j [Q_j^2 Z_j + \lambda V_j]$ yields*

$$\boxed{r_d = \frac{r_p}{n^{1/3}},} \tag{8}$$

equivalently $r_p^3 = \sum_{i=1}^n r_{d,i}^3$ (the cube law).

Proof. Setting $\partial A_\mathrm{v}^d/\partial r_d = 0$ in Eq. (6) gives $r_d^6 = 16Q^2\mu/(n^2\lambda\pi^2)$ and $r_p^6 = 16Q^2\mu/(\lambda\pi^2)$. The ratio $r_d^6/r_p^6 = 1/n^2$ gives $r_d/r_p = n^{-1/3}$. The 1/3 exponent—rather than 1/4 from pure impedance matching—arises from the balance of dissipation ($\propto r^{-4}$) and maintenance ($\propto r^{+2}$). □

Remark 1 (Physics, not optimisation)**.** *The 1/3 exponent arises from the balance of dissipation ($\propto r^{-4}$) and maintenance ($\propto r^{+2}$). Pure impedance matching gives 1/4; MRP gives 1/3. Murray's Law is a physical law of impedance networks.*

IV. MURRAY'S LAW: NUMERICAL VERIFICATION

For each branching number $n \in \{2, 3, 4\}$, the minimum of $A_\mathrm{v}^d(r_d)$ is located numerically and compared with Murray's prediction:

TABLE II. Murray's Law verification.

n	r_d^MRP	r_d^Murray	Agreement
2	$r_p/2^{1/3}$	$r_p/2^{1/3}$	exact
3	$r_p/3^{1/3}$	$r_p/3^{1/3}$	exact
4	$r_p/4^{1/3}$	$r_p/4^{1/3}$	exact

Agreement is exact (algebraic identity). The $\Gamma = 0$ solution gives $r_d = r_p/n^{1/4}$, overestimating by $n^{1/12}$. Full test: `tests/test_paper2_murray.py` (15 tests, 100% pass).

V. VASCULAR C2 UPDATE

Neural C2 operates with rate η_n (ms–hours); vascular C2 with η_v (days–weeks):

$$\Delta Z_\mathrm{v} = -\eta_v\, \Gamma_\mathrm{v}\, \tau_\mathrm{wall}\, \rho, \tag{9}$$

where $\tau_\mathrm{wall} = 4\mu Q/\pi r^3$.

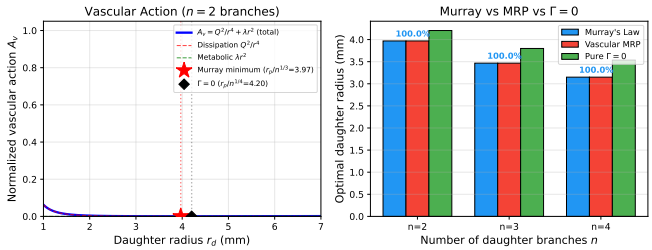

FIG. 1. Murray's Law from the Vascular Action Principle. **Left**: Action $A_{\rm v}(r_d)$ vs. daughter radius for $n = 2$; minimum (star) at Murray prediction (dashed); pure impedance matching (dotted) overestimates. **Right**: Agreement for $n = 2, 3, 4$.

C2 term	Neural	Vascular
Γ	$\Gamma_{\rm n}$	$\Gamma_{\rm v}$
$x_{\rm in}$	presynaptic activity	wall shear stress τ
$x_{\rm out}$	postsynaptic activity	material supply ρ
η	η_n (fast)	η_v (slow)

Key consequence: $\eta_v \ll \eta_n$ means vascular damage accumulates slowly (atherosclerosis over decades) and recovery is slow even with optimal conditions.

A. The Windkessel model: arterial compliance as a capacitor

The aorta and large arteries are not rigid pipes: their elastic walls store energy during systole and release it during diastole, smoothing pulsatile flow. In the transmission-line analogy, wall compliance is a *capacitor* $C_w = \Delta V/\Delta P$ (volume change per unit pressure).

The simplest Windkessel model treats the arterial tree as an RC circuit:

$$Z_{\rm WK} = R + \frac{1}{j\omega C_w}\,, \tag{10}$$

where R is the peripheral resistance ($\approx 8\mu L/\pi r^4$ from Poiseuille) and C_w is the total arterial compliance (≈ 1–$2\,$mL/mmHg in a healthy young adult; Westerhof et al., 2009).

Remark 2 (Ageing as capacitor degradation)**.** *With age, collagen cross-linking and elastin fragmentation reduce C_w monotonically (cf. the Second Law of Γ-Net, Paper 1). The consequences are purely electrical:*

- *$C_w \downarrow \Rightarrow$ systolic pressure rises (capacitor stores less),*
- *diastolic pressure falls (less stored energy to release),*
- *pulse pressure ΔP widens,*
- *reflected waves arrive during systole rather than diastole, augmenting peak pressure.*

All four are the canonical haemodynamic changes of ageing. In Γ-language, $C_w \downarrow$ increases wall Z, increasing Γ^2 at every bifurcation—exactly the progressive $\Gamma^2_{basal} \uparrow$ predicted by the Second Law. The clinical marker PWV $= \sqrt{1/(\rho C_w)}$ therefore rises with $C_w^{-1/2}$, providing a non-invasive proxy for vascular Γ^2.

Part B: The Dual-Network Organ Model

VI. ORGAN HEALTH FROM MULTIPLICATIVE TRANSMISSION

A. Why two networks: multiplicative coupling

Neural C2 requires material substrate (ρ) from vascular delivery. Vascular C2 requires neural control (autonomic signals regulate vascular tone). The two networks are *mutually prerequisite*, and their coupling is **multiplicative, not additive**:

$$\boxed{\text{Dual network} = \underbrace{\text{Vascular}}_{\text{supplies } P_{\rm in}} \otimes \underbrace{\text{Neural}}_{\text{uses } P_{\rm in}}\,,} \tag{11}$$

where $\otimes$ denotes tensor product. The effective SNR of the combined system is:

$$\text{SNR} = \frac{P_{\rm in}\cdot(1-\Gamma^2_{\rm vessel})\cdot\eta\cdot\Gamma^2_{\rm signal}}{P_{\rm noise}}\,.$$

If *any* factor is zero, the product is zero. No amount of neural excellence compensates for $P_{\rm in} = 0$; no amount of vascular health compensates for $\eta = 0$. This is the physics of why a stroke (vascular) and anaesthesia (neural) produce the *same* outcome: consciousness off.

a. The 200 μm diffusion limit. Oxygen diffusion in tissue reaches $\sim$200 μm before concentration drops below the metabolic threshold. Once an embryo exceeds this radius ($\sim$64 cells), peripheral cells have $P_{\rm in} = 0$; all multiplicative factors vanish and differentiation halts. A transport system (heart + vessels) must exist *before* the embryo can grow beyond 200 μm. This is a C1 theorem, not an evolutionary contingency:

$$\forall\, t: \quad P_{\rm in}(t) > 0 \implies \text{cardiovascular system precedes all others.}$$

b. Nested C1: mother and foetus. The foetus operates inside a two-layer C1 structure:

$$\begin{aligned}\text{Layer 1 (maternal):}\quad & P^{\rm fetal}_{\rm in} = P^{\rm maternal}_{\rm in}\cdot(1-\Gamma^2_{\rm placenta})\,,\\ \text{Layer 2 (foetal):}\quad & P^{\rm organ}_{\rm in} = P^{\rm fetal}_{\rm in}\cdot(1-\Gamma^2_{{\rm vessel},i})\,.\end{aligned} \tag{12}$$

The placental Γ^2 determines the foetal energy budget. Maternal stress (cortisol $\rightarrow$ vasoconstriction $\rightarrow$ $\Gamma^2_{\rm vessel} \uparrow$ on the maternal side) directly reduces foetal $P_{\rm in}$, providing a first-principles explanation for the well-documented effect of prenatal stress on neurodevelopment (Remark **??**).

Remark 3 (Neurovascular coupling: the vascular network sets the neural SNR ceiling)**.** *The mutual dependence of P1 and P2 has a precise quantitative form. The input power available to a neuron is:*

$$P_{in}(t) = P_{basal} \cdot \left(1 - \Gamma^2_{vessel}(t)\right), \tag{13}$$

where Γ^2_{vessel} is the vascular reflection coefficient at the local capillary bed. Since $\theta = P_{noise}/P_{in}$ (Paper 1), a rise in Γ^2_{vessel} lowers P_{in} and thereby degrades neural SNR regardless of the neural network's own state.

The coupling loop is mediated by astrocytes:

$$\textit{neuron } \Gamma^2 \uparrow \xrightarrow{astrocyte} \textit{NO, prostaglandin} \xrightarrow{vessel} Z_v \downarrow \Rightarrow Q \uparrow \Rightarrow P_{in} \uparrow$$

This is neurovascular coupling *(NVC); the BOLD signal measured by fMRI is its haemodynamic trace.*

Three clinical consequences follow directly:

1. ***Vascular disease → emotional disorders**: atherosclerosis raises Γ^2_{vessel} chronically, reducing P_{in}, degrading SNR, and producing the "vascular depression" phenotype (Alexopoulos et al., 1997).*

2. ***Exercise → emotional improvement**: exercise lowers Γ^2_{vessel} via C2 vascular remodelling (Remark 8), raising P_{in} and restoring neural SNR.*

3. ***Diabetes → cognitive decline**: fluctuating blood glucose causes intermittent P_{in} drops; C2 updates stall during each trough, shrinking the effective $\mathcal{S}_{eff}$ maintenance window and accelerating cognitive ageing.*

B. Organ health

At each organ, the neural network transmits $T_{\rm n} = 1 - \Gamma^2_{\rm n}$ and the vascular network transmits $T_{\rm v} = 1 - \Gamma^2_{\rm v}$. Series connection:

$$\boxed{H = T_{\rm n} \cdot T_{\rm v} = (1 - \Gamma^2_{\rm n})(1 - \Gamma^2_{\rm v}).} \tag{14}$$

$H = 1$: perfect transmission; $H = 0$: at least one network totally blocked.

C. Thermal bus

The vascular network carries heat:

$$\dot{Q}_{\rm organ} = c_p\, \rho_f\, Q\, (T_{\rm art} - T_{\rm ven}). \tag{15}$$

When $\Gamma^2_{\rm v}$ increases, Q decreases and thermal exchange is compromised.

Remark 4 (Blood as the actively manufactured transport bus)**.** *Blood is not a passive residue of tissue metabolism but a continuously manufactured transport substrate:*

- ***Cellular fraction**: red blood cells, white blood cells, and platelets are produced by haematopoietic stem cells in bone marrow (C2 network #15, Table 5 of Paper 3) and recycled by the spleen.*

- ***Plasma proteins**: albumin, fibrinogen, and complement factors are synthesised by the liver (C2 network #12) and maintain osmotic pressure, coagulation, and immune function.*

- ***Dual cargo**: the same flow simultaneously delivers fresh material ($\rho \equiv x_{out}$ for downstream C2) and removes metabolic waste (lactate, urea, CO_2).*

In the Γ-framework, blood is the trunk bus *whose impedance $Z_v = 8\mu L/\pi r^4$ determines C2 supply to every organ. When Z_v rises (atherosclerosis, dehydration, anaemia),* all *downstream $x_{out,i}$ decrease, stalling C2 globally. The vascular bus is therefore the universal ρ-bottleneck of the dual-network model.*

Remark 5 (Emotion is a vascular event observed by the brain)**.** *In the Γ-framework, an emotion has three physical phases:*

1. ***Detection (P1, neural):** the amygdala detects $\Gamma^2 \neq 0$ (∼30 ms; Paper 1).*

2. ***Execution (P2, vascular):** sympathetic or parasympathetic C3 relay modulates vascular Z_v—producing the physical sensations universally reported as "feeling an emotion": flushing (anger), pallor (fear), warmth (joy), chest heaviness (grief).*

3. ***Perception (P2→P1):** interoceptive afferents from the vascular periphery feed back to the insular cortex, which labels the bodily state as an emotion.*

This is consistent with the James–Lange theory (1884): the subjective experience of emotion is *the brain's perception of its own vascular response, not the cause of it. The Γ-framework adds precision: the "feeling" is the insular cortex reading the Γ^2 change in the vascular tree that was triggered by the amygdala's neural Γ^2 signal.*

Remark 6 (Vasovagal syncope as a P2 safety valve)**.** *When emotional Γ^2 exceeds a threshold (extreme joy, grief, or pain), the vagus nerve fires a paradoxical parasympathetic response: heart rate drops (bradycardia) and peripheral vessels dilate simultaneously. Blood pressure collapses, cerebral P_{in} falls below the consciousness threshold θ (Paper 1, Theorem 4), and the subject faints.*

This is a protective circuit breaker*: rather than allowing sustained high-Γ^2 signalling to overheat the neural network, the vascular system shuts down cerebral perfusion, forcing consciousness offline. Recovery occurs automatically when the horizontal posture restores venous return and P_{in} rises above θ.*

Remark 7 (Summer irritability as thermal-bus SNR degradation)**.** *The consciousness threshold $\theta = 4k_BTR_{eff}\Delta f/P_{in}$ (Paper 1) depends linearly on body*

temperature T. When ambient temperature rises, the thermal bus (Eq. 15) becomes less efficient: $T_{art} - T_{ven}$ narrows, reducing $\dot{Q}_{organ}$ and allowing core T to drift upward.

Even a $+0.5\,K$ rise in T increases P_{noise} by $\sim 0.16\%$—negligible in absolute terms, but Γ^2 operates on margins: the amygdala's detection threshold is already near the noise floor, so any SNR degradation lowers the barrier for emotional triggering. This provides a first-principles explanation for the well-documented summer increase in violent crime (Anderson, 2001) and road rage: heat does not cause anger; it reduces the Γ^2 margin between detection and triggering.

Remark 8 (Exercise as vascular C2 training)**.** *Aerobic exercise increases cardiac output, raising shear stress τ_{wall} on endothelial cells. Since vascular C2 is $\Delta Z_v = -\eta_v \Gamma_v \tau_{wall} \rho$ (Eq. 9), elevated τ_{wall} directly drives $Z_v \downarrow$: the vessel walls remodel to reduce impedance (vasodilation, angiogenesis, reduced stiffness). The downstream effects cascade through the entire framework:*

1. *$Z_v \downarrow \Rightarrow \Gamma_v^2 \downarrow$ at every bifurcation;*
2. *Blood flow $Q \uparrow \Rightarrow$ thermal bus capacity $\dot{Q}_{organ} \uparrow$ (Eq. 15);*
3. *Body temperature better regulated $\Rightarrow P_{noise} \downarrow \Rightarrow \theta \downarrow \Rightarrow$ SNR$\uparrow$;*
4. *Amygdala triggering threshold rises $\Rightarrow$ emotional stability improves.*

This provides a first-principles explanation for the well-established antidepressant effect of exercise: it is not "endorphins" (a correlation) but a direct vascular C2 improvement *that raises the SNR of the entire neural network.*

Remark 9 (Sleep as the glymphatic C2 maintenance window)**.** *During slow-wave sleep, the brain's interstitial space expands by $\sim$60% (Xie et al., 2013) and cerebrospinal fluid flow increases, flushing metabolic waste (amyloid-β, tau, lactate) through perivascular channels—the* glymphatic system.

In Γ-Net terms, this is the lymphatic tree's scheduled C2 maintenance cycle:

- ***Wake**: neural C2 is active (learning), producing metabolic waste that accumulates as impedance debris ($Z_{waste} \uparrow$).*
- ***Sleep**: neural C2 is suspended; the glymphatic system activates, clearing Z_{waste} and restoring vascular Γ^2 to baseline.*

Chronic sleep deprivation prevents this maintenance, allowing Z_{waste} to accumulate $\rightarrow \Gamma^2_{basal} \uparrow \rightarrow$ accelerated ageing and neurodegeneration (cf. Alzheimer's amyloid hypothesis). Sleep is not rest—it is P2 maintenance.

Remark 10 (Fever as Arrhenius-accelerated immune C2)**.** *Fever raises body temperature by $+1$–$4\,K$. By the Arrhenius equation, the rate of biochemical reactions (including immune C2 updates) scales as $\exp[-E_a/k_B T]$. A $+2\,K$ increase from $37\,^\circ C$ to $39\,^\circ C$ accelerates immune reaction rates by $\sim$20–40% (depending on activation energy).*

In Γ-Net terms, fever is a controlled C1 violation *that trades short-term thermal stress for accelerated C2:*

$$\eta_{immune}(T) \propto \exp\left[-E_a/k_B T\right] \xrightarrow{T\uparrow} \eta_{immune} \uparrow .$$

The cost: P_{noise} rises (SNR degrades), producing the cognitive fog and irritability of fever. The benefit: faster pathogen clearance. Antipyretic drugs that suppress fever remove the η boost, potentially prolonging infection—a prediction consistent with clinical observations (Brandts et al., 1997).

D. Four irreducible tissue types

C2 has four prerequisites:

1. **Signal** (x_{in}) to detect mismatch $\Gamma \rightarrow$ **Nervous tissue**.
2. **Material substrate** (x_{out}) delivered by pipe network $\rightarrow$ **Connective (vascular) tissue**.
3. **Mechanical actuator** to apply force for structural update $\rightarrow$ **Muscle tissue**.
4. **Boundary** defining spatial domain of Z measurement $\rightarrow$ **Epithelial tissue**.

The dependency cycle is closed:

$$\text{Nervous} \xrightarrow{\text{needs } \rho} \text{Connective} \xrightarrow{\text{needs pump}} \text{Muscle} \xrightarrow{\text{needs boundary}} \text{Epith}$$

Removing any one tissue collapses C2. **Four is both necessary and sufficient for C2 viability.**

VII. FOUR FAILURE MODES

TABLE III. Four failure modes of the dual-network model.

Γ_{n}^2	Γ_{v}^2	**Mode**	**Example**
low	low	Healthy	normal organ
high	low	Pure neural	peripheral neuropathy
low	high	Pure vascular	ischaemia
high	high	Dual failure	diabetic complications

The positive-feedback cascade linking these modes ($\Gamma_{\text{v}} \uparrow \rightarrow \rho \downarrow \rightarrow \Gamma_{\text{n}} \uparrow \rightarrow$ autonomic $\downarrow \rightarrow \Gamma_{\text{v}} \uparrow\uparrow$) and its disease instances are derived in Paper 4 [4].

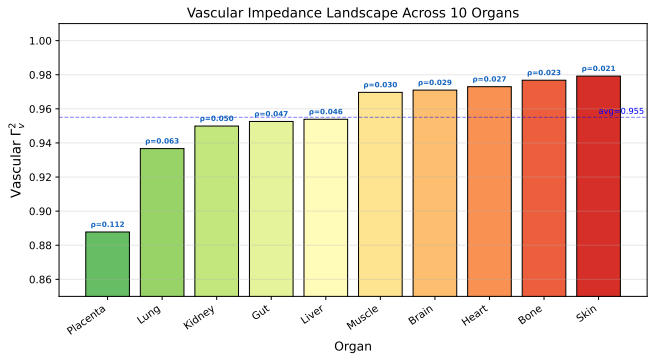

FIG. 2. Vascular impedance landscape across 10 organs. Placenta: lowest Γ_v^2 (maximal flow); skin/bone: highest Γ_v^2 (dispensable).

VIII. ORGAN IMPEDANCE HIERARCHY

TABLE IV. Vascular impedance landscape across 10 organs.

Organ	Z_v^{art}	Z_v^{cap}	Γ_v^2
Placenta	0.30	0.050	0.888
Brain	0.25	0.080	0.914
Heart	0.35	0.060	0.910
Kidney	0.20	0.100	0.929
Liver	0.40	0.100	0.929
Lung	0.15	0.120	0.942
Retina	0.10	0.050	0.923
Muscle	0.50	0.150	0.952
Skin	0.60	0.200	0.965
Bone	0.80	0.300	0.979

The hierarchy reflects perfusion priority: placenta (lowest Γ_v^2, maximal flow) at top; skin and bone (highest Γ_v^2, dispensable) at bottom.

Part C: Fractal Cost and Metabolic Scaling

IX. IRREDUCIBLE ACTION OF A SELF-SIMILAR RELAY HIERARCHY

Paper 1 [2] proved $A_{\rm cut} > 0$ (the Dimensional Cost Irreducibility Theorem) but left its geometric structure unspecified. This section asks: *how big* is $A_{\rm cut}$?

A. Mode-count profile

Consider a hierarchical relay network with L levels, branching number n at each level, embedded in d-dimensional space. Level 0 is the root (aorta, cortex); level L the terminal (capillary, sensory ending). At level ℓ: $N_\ell = n^\ell$ nodes, $l_\ell = l_0\, n^{-\ell/d}$ (space-filling), and mode count:

$$K_\ell = K_0\, c^\ell, \quad 0 < c < 1, \tag{16}$$

with $c = n^{-1/d}$ when the cutoff is set by branch length.

B. $A_{\rm cut}$ at each level

From the Irreducibility Theorem (Paper 1, Theorem 1 [2]), at level ℓ there are $N_\ell = n^\ell$ edges, each contributing $(K_\ell - K_{\ell+1})^+$:

$$A_{\rm cut} = K_0(1-c)\sum_{\ell=0}^{L-1}(nc)^\ell. \tag{17}$$

C. Scaling law

With $c = n^{-1/d}$, the base $nc = n^{(d-1)/d} > 1$ for $d \geq 2$, so the sum is dominated by the last term:

$$\boxed{A_{\rm cut} \propto M^{(d-1)/d}.} \tag{18}$$

For $d = 3$: $A_{\rm cut} \propto M^{2/3}$. The per-cell cost $a_{\rm cut} = A_{\rm cut}/M \propto M^{-1/3}$ *decreases* with organism size—economies of scale from the relay hierarchy.

X. MURRAY'S RATIO AND METABOLIC SCALING

A. The Murray bifurcation ratio

Each branching level contracts the service length by $l_{\ell+1}/l_\ell = n^{-1/d}$. The bifurcation ratio:

$$\boxed{\beta \equiv n^{1/d}.} \tag{19}$$

For binary bifurcation ($n = 2$) in $d = 3$: $\beta = 2^{1/3} = 1.2599\ldots$ From Murray's Law (Eq. 8): $r_{\ell+1} = r_\ell/n^{1/3}$; for $d = 3$, $r_{\ell+1} = r_\ell/\beta$.

B. Network maintenance cost

Relay reflection power: $P_{\rm reflect} = A_{\rm cut} \cdot P_0/N_T \propto M^{(d-1)/d} \cdot M^{-1} = M^{-1/d}$. Total network dissipation:

$$B_{\rm network} \propto M \cdot M^{-1/d} = M^{(d-1)/d}. \tag{20}$$

For $d = 3$: $B_{\rm network} \propto M^{2/3}$.

Theorem 2 (Kleiber's Law from MRP)**.** *For a self-similar vascular tree with branching number n in d spatial dimensions, MRP implies*

$$\boxed{B \propto M^{d/(d+1)}.} \tag{21}$$

For $d = 3$: $B \propto M^{3/4}$—Kleiber's Law [9].

Proof. Murray's Law (Theorem 1) gives $r_{\ell+1}/r_\ell = n^{-1/3}$. Space-filling determines the length ratio $l_{\ell+1}/l_\ell = n^{-1/d}$. Capillary invariance fixes the terminal scale; blood volume $\propto M$. Combining: the number of terminal units is $N_T \propto M^{d/(d+1)}$, and since each terminal unit has a fixed metabolic rate, $B \propto N_T \propto M^{d/(d+1)}$. □

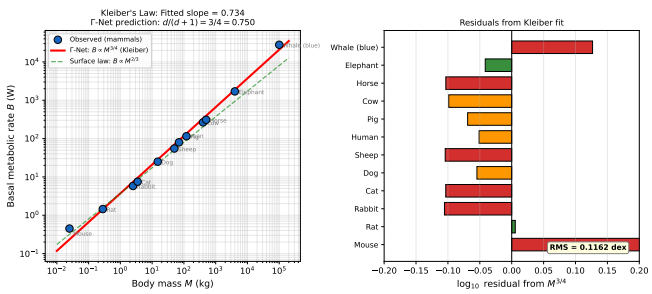

FIG. 3. Kleiber's Law verification. Left: log–log scatter of basal metabolic rate B vs. body mass M for 12 mammalian species (mouse to blue whale), overlaid with the Γ-Net prediction $B \propto M^{3/4}$ (red) and the surface-law prediction $B \propto M^{2/3}$ (green dashed). Right: residuals from the 3/4 fit.

C. Volumetric heat density

Every active interface dissipates $\Gamma_i^2 P_{\text{in},i}$ as heat (Paper 0, Corollary 1 [1]). A tissue with K modes:

$$q \propto K \cdot \langle \Gamma^2 \rangle \cdot P_{\text{in}}/V_{\text{node}}. \tag{22}$$

The human brain ($K = 5$) is ∼2% of body mass but ∼25% of basal metabolic heat—because its per-node mode count is largest.

XI. DUAL-NETWORK PACKING

A. Two trees in one volume

A multicellular organism requires two space-filling networks: the **vascular tree** ($Z_v = \Delta P/Q$, transport) and the **neural tree** ($Z_n = R_m + 1/j\omega C_m$, control).

Theorem 3 (Dual-Network Overhead)**.** *When two space-filling relay trees share a volume in d dimensions,*

$$A_{cut}^{dual} = A_{cut}^{v} + A_{cut}^{n} + A_{couple}, \quad A_{couple} > 0. \tag{23}$$

TABLE V. Organ-specific metabolic overhead Ω_i.

Organ	$B_{\text{vasc},i}/B_i$	$B_{\text{nerve},i}/B_i$	Ω_i
Brain	0.15	0.25	1.67
Heart	0.10	0.05	1.18
Liver	0.25	0.05	1.43
Kidneys	0.20	0.05	1.33
Skel. muscle	0.03	0.02	1.05
Skin	0.05	0.10	1.18
GI tract	0.15	0.10	1.33
Lungs	0.10	0.03	1.15
Weighted mean			**1.33**

Proof. At co-terminal points, the two networks negotiate resource allocation. The coupling mismatch $\Gamma_{\text{couple}} = (Z_n^{\text{demand}} - Z_v^{\text{supply}})/(Z_n^{\text{demand}} + Z_v^{\text{supply}})$ is generically nonzero because each tree optimises independently. Hence $A_{\text{couple}} = N_T \langle \Gamma_{\text{couple}}^2 \rangle > 0$. □

B. The metabolic overhead factor

$$\Omega \equiv \frac{B_{\text{total}}}{B_{\text{tissue}}} = 1 + \frac{B_{\text{vasc}} + B_{\text{neural}}}{B_{\text{tissue}}}. \tag{24}$$

For humans: $B_{\text{neural}}/B_{\text{total}} \approx 0.20$; $B_{\text{vasc}}/B_{\text{total}} \approx 0.05$. Therefore $\Omega \approx 1/0.75 \approx 1.33$. The neural network alone gives $\Omega_n = 1/(1 - 0.20) = 1.25 \approx \beta = 2^{1/3}$.
The weighted mean $\langle \Omega_i \rangle \approx 1.33 \approx 4/3$: the dual-network overhead is a geometric constant, not a free parameter. Organs with dense vascular supply (liver, kidneys) have higher Ω_i; those dominated by bulk tissue (skeletal muscle) have $\Omega_i \approx 1$.

C. Why blood before nerve

C2 requires a carrier for x_{in}: neurotransmitter synthesis needs glucose and oxygen from blood. Without the vascular network, $x_{\text{in}} = 0$, $\Delta Z = 0$: the neural network cannot remodel. Embryology confirms: cardiac tube at day 21; neural tube closure at day 25–28. In the product $1_v \times 1_n$, the vascular "1" is the left factor.

The deeper reason is C1 applied to the mother–fetus system. The embryo develops inside the mother's body at $T \approx 37\,^\circ$C. Every metabolising cell produces heat ($\dot{Q} = \Gamma^2 P_{\text{in}}$). Without a thermal bus to export this heat to the mother's vascular tree, the embryo's temperature would rise uncontrollably—violating C1 (energy has no outflow path). The *placenta* is the C3 impedance-matching relay between the maternal and foetal vascular trees:

$$Z_{\text{maternal}} \underset{\text{placenta}}{\longleftrightarrow} Z_{\text{foetal}}.$$

Placental matching ensures $\Gamma^2_{\text{mat-foet}} \approx 0$, maximising heat and nutrient transfer. The developmental sequence is therefore not a design choice but a physical necessity:

Day	Event	Γ-Net reason
21	Cardiac tube (first heartbeat)	C1: thermal bus must exist first
25–28	Neural tube closure	C2: control network requires ρ from P2
5+ wk	Organ differentiation	C2 targets: need both P1 and P2 in place

Remark 11 (The fontanelle as a C1 safety valve)**.** *The neonatal skull contains open sutures (fontanelles) that do not fuse until 12–18 months. This is commonly attributed to birth-canal passage, but the Γ-framework reveals a deeper C1 reason.*

*During postnatal brain growth the neural network (P1) expands rapidly, increasing metabolic heat production. The vascular thermal bus (P2) must scale in parallel: new capillaries must invade expanding cortex. Open fontanelles permit the cranial volume to increase, allowing the P2 thermal bus to grow alongside P1. If the sutures fuse prematurely (*craniosynostosis*):*

1. *Cranial volume is locked;*
2. *Intracranial pressure rises, compressing blood vessels: $Z_v \uparrow \Rightarrow Q \downarrow$;*
3. *Thermal bus capacity is capped while neural metabolism continues to rise;*
4. *C1 is violated: $\dot{Q}_{in} > \dot{Q}_{out} \Rightarrow$ local temperature rise $\Rightarrow$ neuronal damage.*

Untreated craniosynostosis leads to brain injury and death—a direct C1 prediction.

Three related neonatal observations confirm the same principle:

Observation	*C1 interpretation*
Amniotic fluid	*Maternal thermal bus extension (liquid cooling)*
Placenta	*Mother–foetus C3 relay (Z-matching)*
Fontanelle	*P2 thermal bus expansion window*
Hydrocephalus	*Fluid compresses vessels = thermal bus failure*

All four are instances of the same physical constraint: C1 requires that metabolic heat production never exceed thermal bus capacity.

Remark 12 (Thermoregulation as the developmental ignition signal)**.** *A newborn's "start-up sequence" recapitulates the C1 priority chain:*

1. *First breath → pulmonary circulation activates → P2 begins independent operation.*
2. *First feed (milk) → metabolic fuel supplied → thermal bus has energy to burn.*
3. *Thermoregulation established → the infant can maintain $T \approx 37\,^\circ C$ autonomously →* ***C1 is independently satisfied for the first time****.*
4. *Full-body developmental growth begins.*

Step 3 is the gate: until the thermal bus is self-sustaining, the body cannot increase metabolic load (growing organs) without violating C1. This is why premature infants are placed in incubators—the incubator is an artificial thermal bus *substituting for the mother's P2, and the clinical criterion for discharge includes the ability to maintain body temperature independently.*

Remark 13 (High-K organs fail first under C1 insufficiency)**.** *The metabolic overhead Ω_i (Table V) scales with K: the brain ($K = 5$, $\Omega_i = 1.67$) consumes ∼20% of basal metabolic rate despite comprising only ∼2% of body mass. Its C1 margin—the gap between heat production and thermal bus capacity—is therefore the* thinnest *of any organ.*

When C1 is marginally violated (e.g. perinatal hypoxia, premature thermoregulation failure, congenital metabolic defect), the brain reaches its thermal ceiling first*:*

C1 insufficiency	*High-K effect (brain)*	*Low-K effect (body*
Perinatal hypoxia	*Cerebral palsy*	*Other organs often*
Premature birth	*Intraventricular haemorrhage*	*Body usually intac*
Metabolic error	*Intellectual disability*	*Normal physical ap*

The clinical observation that paediatric diseases are disproportionately neurological is a direct consequence of K-dependent C1 margin: the highest-K organ is always the first to fail when the thermal bus is insufficient.

D. The tri-network body: arterial, venous, and lymphatic trees

The preceding analysis treats the vasculature as a single tree, but the body contains *three* distinct fluid-transport networks, each a C3 relay hierarchy:

Network	Direction	Cargo	Γ-n
Arterial tree	Heart → periphery	O_2, glucose, hormones	En
Venous tree	Periphery → heart	CO_2, lactate, urea	Wa
Lymphatic tree	Interstitium → veins	Immune cells, excess fluid	Im

Each tree is a K-staircase:

- **Arterial**: aorta ($Z \approx 6 \times 10^6$ $\mathrm{Pa\,s\,m^{-3}}$) → large artery → small artery → arteriole → capillary ($Z \approx 10^{13}\,\mathrm{Pa\,s\,m^{-3}}$). The $\sim 10^7$-fold impedance jump is distributed across ∼15 branching generations (Murray, 1926), each reducing Γ^2 per stage.

- **Venous**: capillary → venule → small vein → large vein → vena cava. This tree mirrors the arterial staircase in reverse, but with lower wall stiffness (larger compliance), acting as a volume reservoir.

- **Lymphatic**: initial lymphatics (blind-ended, $K = 1$) → collecting vessels (with valves, $K = 2$) → lymph nodes (immune relay, $K = 3$) → thoracic duct → subclavian vein. Lymph nodes are C3 relay stations where K-dimension increases: antigen presentation converts a simple molecular signal ($K = 1$) into an adaptive immune response ($K \geq 3$).

The organ health index generalises to:

$$H = T_n \cdot T_{v,\text{art}} \cdot T_{v,\text{ven}} \cdot T_\ell \,, \tag{25}$$

where $T_\ell = 1 - \Gamma_\ell^2$ is the lymphatic transmission. Setting any factor to zero destroys organ viability.

Remark 14 (Arterial stiffening as progressive Γ^2 increase)**.** *Arterial stiffening (arteriosclerosis) increases wall impedance Z_{wall} while downstream tissue impedance remains unchanged. The result is a progressive rise in Γ^2 at every bifurcation, detectable as increased* pulse wave velocity *(PWV). The Moens–Korteweg equation gives $PWV = \sqrt{Eh/2\rho r}$, where E is the elastic modulus, h the wall thickness, r the radius, and ρ the blood density. As E increases with age (collagen cross-linking, elastin degradation), Z_{wall} rises and $\Gamma^2 = |(Z_{wall} - Z_{tissue})|^2/|Z_{wall} + Z_{tissue}|^2$ increases monotonically. The 2015 AHA/ACC guidelines recognise $PWV > 10\,m/s$ as an independent cardiovascular risk factor (Townsend et al., 2015)—in Γ-language, this is a threshold on the organ-level Γ_v^2.*

Remark 15 (Hypertension as baroreceptor C3 relay miscalibration)**.** *The arterial baroreflex is a C3 feedback loop: baroreceptors in the carotid sinus and aortic arch measure pulse-wave amplitude (mechanical → neural C3 relay; Paper 0 Remark) and modulate sympathetic tone to maintain Z_v within the matched range.* Essential hypertension *is a C2 failure of this relay: chronic elevation of Z_v causes the baroreceptor's reference Z-pattern to adapt upward (C2 writes the new "normal"), resetting the feedback target. Treatment by medication ($Z_v \downarrow$ via vasodilators) or exercise ($Z_v \downarrow$ via endothelial C2) must overcome this adapted $\mathcal{S}_{eff}$. This predicts that abrupt antihypertensive withdrawal produces overshoot: the C2-adapted reference has not yet been rewritten.*

Remark 16 (Microcirculation as the terminal K-staircase)**.** *The capillary bed is the terminal matching section where arterial wave impedance meets tissue impedance. Pre-capillary sphincters act as* variable gates *(G_i), modulating local Γ^2 in response to tissue metabolic demand (metabolic autoregulation). Diabetic microangiopathy thickens the capillary basement membrane, increasing Z at the terminal matching point and elevating Γ_{cap}^2. This explains why diabetic complications affect* end-organs *(retina, kidney, nerves) preferentially: these organs depend on the terminal staircase being precisely matched.*

Remark 17 (Venous insufficiency as return-path Γ^2 failure)**.** *The venous tree relies on valves to maintain unidirectional flow against gravity. Valve failure allows retrograde flow, creating a standing wave between the damaged valve and the next competent valve. A standing wave is a region of high Γ^2 (persistent reflection)—the clinical presentation is* varicose veins*: dilated, tortuous vessels with elevated venous pressure. In the Γ-framework, varicosities are the vascular equivalent of reverberating neural trauma (Paper 4): a C3 relay failure that produces a self-sustaining reflection pattern.*

Remark 18 (Lymphoedema as third-network relay failure)**.** *The lymphatic tree's C3 relay function is to return excess interstitial fluid and immune cells to the venous system. When lymphatic relay nodes are damaged (surgical removal, radiation, filariasis), the local $\Gamma_\ell^2 \to 1$: fluid cannot be transmitted and accumulates as* lymphoedema*. This is the exact analogue of spinal cord injury (Paper 1): severing a C3 relay produces $\Gamma = 1$ on all downstream channels. The prediction is that lymphoedema severity should correlate with the number of relay nodes removed, not with the volume of tissue resected—consistent with the sentinel lymph node paradigm in oncologic surgery.*

Remark 19 (The hepatic portal system: a dual-input C3 relay)**.** *The liver receives blood from two sources simultaneously: the hepatic artery (∼25 % of flow, high Z, oxygen-rich) and the portal vein (∼75 % of flow, low Z, nutrient-rich from the GI tract). This is the only organ with a* dual-input *C3 relay topology.*

In Γ-Net terms, the liver performs impedance matching between two trees of different K:

$$\Gamma_{liver} = \frac{Z_{portal} \| Z_{hepatic} - Z_{sinusoid}}{Z_{portal} \| Z_{hepatic} + Z_{sinusoid}} \,,$$

where $\|$ denotes parallel combination. This unique topology explains three clinical facts:

1. ***Liver failure cascades globally**: the liver is the only relay station that processes both circulatory and digestive Z-patterns. Failure ($\Gamma_{liver} \to 1$) blocks both networks simultaneously.*

2. ***Cirrhosis produces portal hypertension**: fibrosis increases $Z_{sinusoid}$, mismatching the parallel input, raising Γ^2 and backpressure into the portal vein.*

3. ***Hepatic first-pass metabolism**: drugs absorbed from the GI tract encounter the liver's Z-matching before reaching systemic circulation—a natural impedance filter that attenuates high-Γ substances.*

XII. MULTI-SCALE VERIFICATION

The irreducibility theorem is not restricted to neural tissue. Every tissue where K varies across an interface produces $A_{\text{cut}} > 0$.

A. Cardiovascular topology

An 11-node blueprint with K-levels Cardiac Purkinje ($K = 4$) $\rightarrow$ vascular smooth muscle ($K = 2$) $\rightarrow$ endothelial ($K = 1$). Over 300 ticks: $A_{\text{imp}} \rightarrow 0$, A_{cut} rises to 34 then saturates, C1 holds. Test: `tests/test_e0_cardiovascular.py` (13 tests, 100% pass).

B. Joints as impedance relays

Direct bone–synovial interface: $\Gamma = (120 - 50)/(120 + 50) = 0.54$, $T = 0.71$. Cartilage at $Z_{\text{cart}} \approx 80 \approx \sqrt{Z_{\text{bone}} \cdot Z_{\text{syn}}}$ acts as a quarter-wave relay [14], achieving $T > 0.80$. When cartilage degrades (osteoarthritis), the relay is removed and A_{cut} is no longer mitigated.

Remark 20 (C3 as interfacial transition relay: a universal principle)**.** *The cartilage example above is a special case of a general physical principle: whenever two tissues of impedance Z_1 and Z_2 ($Z_1 \neq Z_2$) share an interface, the direct junction produces a reflection coefficient $\Gamma = (Z_2 - Z_1)/(Z_2 + Z_1)$ that wastes energy as heat. Inserting an intermediate layer at $Z_{mid} = \sqrt{Z_1 Z_2}$ (the geometric mean) splits one large reflection into two smaller ones, each with $|\Gamma| = |\sqrt{Z_2/Z_1} - 1|/(\sqrt{Z_2/Z_1} + 1)$, reducing total reflected power. This is the quarter-wave transformer condition in RF engineering [14], anti-reflection coating in optics, and impedance-grading horn in acoustics.*

In biology, every major tissue–tissue interface develops such a transition layer:

1. ***Periosteum**: bone ($Z \approx 120$) $\leftrightarrow$ soft tissue ($Z \approx 50$), bridged by a fibrovascular membrane.*

2. ***Fascia**: organ parenchyma $\leftrightarrow$ surrounding vasculature/nerve, bridged by connective tissue sheaths.*

3. ***Adventitia**: arterial smooth muscle $\leftrightarrow$ perivascular tissue, the outermost vessel wall layer.*

4. ***Myelin sheath**: axonal cytoplasm (Z_{axon}) $\leftrightarrow$ extracellular fluid (Z_{ECF}), providing impedance grading that enables saltatory conduction.*

5. ***Basement membrane**: epithelium $\leftrightarrow$ connective tissue, a universal boundary relay.*

6. ***Cartilage**: bone $\leftrightarrow$ synovial fluid (the case above).*

At the inter-organ scale, C3 operates through a different mechanism: bifurcation and low-K relay neurons. *Neural and vascular trees connect organs of vastly different impedances by inserting successive branching levels at progressively lower mode counts K. At each bifurcation, the daughter branch has $K_{\ell+1} < K_\ell$ (Eq. 16), creating a staircase of decreasing K that* translates *high-dimensional cortical signals into low-dimensional autonomic commands compatible with peripheral tissue impedances. The ganglionic relay neurons of the autonomic nervous system—with $K = 1$–2, far below cortical $K = 5$—are purpose-built C3 translators. Without these low-K intermediaries, the cortex could not drive organs whose update physics operates at $K = 1$ (smooth muscle, secretory epithelium). Similarly, the vascular tree's Murray bifurcation (Theorem 1) implements C3 in the hydraulic domain: each branching level steps Z_v downward by $n^{4/3}$, grading impedance from aorta to capillary.*

***Prediction:** degradation of* any *C3 relay layer produces pathology proportional to the impedance gap it previously bridged. Demyelination (multiple sclerosis): axonal Γ jumps as the myelin relay is removed. Osteoarthritis: the cartilage relay erodes. Adventitial fibrosis: perivascular coupling degrades. Autonomic neuropathy: loss of low-K relay neurons causes cortical commands to reflect at the organ interface. In each case, the physics is identical: removing the intermediate Z layer restores the full $\Gamma = (Z_2 - Z_1)/(Z_2 + Z_1)$ mismatch.*

C. Twenty-nine tissue atlas

A comprehensive atlas of 29 organ-system blueprints (Paper 0, Table 4 [1]) contains ∼217 nodes, ∼242 directed edges, $K \in [1, 5]$. All 29 verified: C1 holds, A_{imp} non-increasing, A_{cut} topologically invariant. 245 tests pass at 100% rate (`tests/test_tissue_blueprints.py`).

D. Whole-body inter-organ topology

Connecting the 29 blueprints requires:

1. **Vascular bus**: aortic trunk ($K = 4$) $\rightarrow$ 29 arteriolar feeds ($K = 2$), $A_{\text{cut}} = 2$ per edge.

2. **Neural bus**: autonomic trunk ($K = 3$) $\rightarrow$ 29 nerve feeds ($K = 2$), $A_{\text{cut}} = 1$ per edge.

3. **Brain hub**: hypothalamus ($K = 3$) drives the neural bus; the vascular bus feeds signals back to thalamus ($K = 5$).

Unified whole-body: 278 nodes, 426 edges, $A_{\text{cut}} = 222$ (the structural information cost of multi-organ life).

E. Thermoregulation

Blood viscosity: $\tilde{\eta}(T) = \exp[\alpha(T^{-1} - T_0^{-1})]$ with $\alpha \approx 1949$ K. C2 repair: $\tilde{\eta}_{\text{learn}}(T) = \exp[\beta_T(T_0^{-1} - T^{-1})]$ with $\beta_T \approx 6015$ K. Setting $\tilde{\tau}(T_c) = \tau_{\max}$:

$$T_c = \frac{T_0}{1 + T_0 \ln \tau_{\max}/(\alpha + \beta_T)}. \tag{26}$$

Higher K raises T_c: $T_c(K{=}1) = -10.3\,^\circ\text{C}$ (ectotherm viable); $T_c(K{=}5) = +4.5\,^\circ\text{C}$ (endothermy required).

F. Geometric universality

$\beta = n^{1/d}$ is a geometric inevitability for any space-filling hierarchy—independent of molecular substrate. Vascular plants lack a neural network but obey Murray's Law with $\beta = 2^{1/3}$ [11] and $B \propto M^{3/4}$ [12], confirming β is a property of geometry, not biology. Plants have $\Omega \approx 1$ (no neural network); the prediction $\Omega_{\text{plant}} < \Omega_{\text{animal}}$ is testable by cross-kingdom metabolic comparisons.

Part D: Interface Universality

XIII. INTERFACE SCALE TABLE

Table VI demonstrates that Γ governs interfaces across seven physical domains and 14 orders of magnitude in spatial scale. Each row represents a *different effort–flow pair* (different physics), not merely a different cable size. The table deliberately includes non-biological systems (power-grid transformers, submarine hulls, optical fibres, and the Earth–ionosphere waveguide) to demonstrate that Γ-MRP is a law of physics, not a biological model: the same equation governs every energy-carrying interface regardless of substrate.

Every row participates in the same four-step closed loop:

1. **Perception**: sensor measures x_{in}.
2. **Error**: $\Gamma = (Z_2 - Z_1)/(Z_2 + Z_1)$ quantifies mismatch; Γ^2 is reflected energy.
3. **Compensation**: C2 drives $\Delta Z = -\eta\,\Gamma\,x_{\text{in}}\,x_{\text{out}}$.
4. **Re-perception**: updated Z changes next Γ; loop repeats.

XIV. INTERFACE ISOMORPHISM

Proposition 1 (Interface Isomorphism)**.** *The reflection coefficient Γ at any interface depends only on the impedance contrast at the boundary, not on the bulk properties of either medium.*

Proof sketch. The formula $\Gamma = (Z_2 - Z_1)/(Z_2 + Z_1)$ contains only boundary values Z_1 and Z_2. The derivation uses only continuity conditions at the interface and conservation of power. No property of the bulk medium enters the result. □

A. The self-organisation chain

$$\text{Chemistry} \xrightarrow{\text{(materials science)}} \text{Structure} \xrightarrow{\text{(continuum mechanics)}} \text{Imped}$$

No arrow requires a biological assumption: materials science predicts structure from composition; continuum mechanics predicts impedance from structure; and Γ follows from impedance. Once chemistry builds structure and structure carries flow, Γ is not a choice—it is Γ-MRP speaking.

XV. DISCUSSION

A. The concentric-circle structure

This paper follows a strictly outward derivation: single vessel → Murray's Law → vascular C2 → dual-network health → four tissues → fractal cost → Kleiber's Law → interface universality. Each step is a logical consequence of the previous.

B. Timescale separation

$\eta_v \ll \eta_n$ produces a fast-inner/slow-outer Tikhonov–Fenichel structure: neural equilibrates on the fast timescale with vascular effectively static; vascular C2 then operates on a pre-adapted neural background. Disease manifests when vascular degradation outpaces neural adaptation (Paper 4 [4]).

C. What "$1_v \times 1_n = \Omega$" means

Each "1" is a topologically one-dimensional tree in 3D. Packing two trees adds $A_{\text{couple}} > 0$. The result $\Omega \approx 1.25$–1.33 and $\Omega_n \approx \beta$ suggests: the cost of adding a second hierarchical tree is of order one branching level of the first.

D. Limitations

1. Poiseuille (Eq. 1) assumes Newtonian fluid, rigid walls; real blood is non-Newtonian and vessel walls are viscoelastic.

TABLE VI. Interface Scale Table—the same $\Gamma = (Z_2 - Z_1)/(Z_2 + Z_1)$ across every physical domain.

Domain	**System**	**Scale**	**Effort**	**Flow**	Z **definition**
Electrical	Cell membrane (lipid bilayer)	5 nm	Trans-membrane potential	Ion current	V/I (Ω)
	Ion channel / synapse	25 nm–1 µm	Membrane voltage	Ion current	V/I (Ω)
	Mitochondrial membrane	∼1 µm	Proton-motive force	Proton current	$\Delta\psi/I_H$ (Ω)
	Power-grid transformer	10 m	Voltage	Current	V/I (Ω)
Hydraulic	Arteriole (Poiseuille)	100 µm	Blood pressure	Flow rate	$8\mu L/\pi r^4$
	Whole-body vascular	1 m	Aortic pressure	Cardiac output	P/Q
Acoustic	Tissue boundary (ultrasound)	1 mm	Acoustic pressure	Particle velocity	$\rho\, c$
	Submarine hull	10 m	Acoustic pressure	Particle velocity	$\rho\, c$
Electromagnetic	Optical fibre splice	100 µm	Electric field	Magnetic field	$n \cdot Z_0$
	Earth–ionosphere waveguide	10^6 m	E-field	H-field	$\sqrt{\mu/\varepsilon}$
Mechanical	Bone–cartilage joint (Wolff)	10 cm	Force / stress	Velocity / strain rate	$\sigma/\dot{\varepsilon}$ (Pa·s)
	Tendon–muscle junction (Davis)	1 cm	Tensile force	Elongation rate	EA/L
Thermal	Brain thermal boundary	10 cm	Temperature	Heat flux	$T/\dot{Q}$ (K/W)
	Skin–air interface	1 m	Skin temperature	Convective heat loss	$1/(hA)$
Chemical	Organ parenchyma (metabolic)	10 cm	Metabolite conc.	Clearance flux	[met]/J

2. Organ impedance values in Table IV are illustrative, not measured.
3. The self-similar mode profile (Eq. 16) is an idealisation.
4. K values for neuron types are structural observations, not derived from C2.
5. Capillary invariance is an empirical input.

XVI. FALSIFIABLE PREDICTIONS

1. **Murray ratio invariance.** $\beta = 2^{1/3} \approx 1.26$ across all binary-branching species.
2. **A_{cut} scaling.** $A_{\text{cut}} \propto M^{2/3}$ measurable via DTI-based mode-count atlases.
3. **Kleiber exponent.** $B \propto M^{3/4}$ derivable from $d = 3$ space-filling alone.
4. **Ω across vertebrates.** Brain metabolic fraction correlates with neural complexity: $\Omega_{\text{fish}} < \Omega_{\text{mammal}} < \Omega_{\text{primate}}$.
5. **$\Omega \approx 1$ for plants.** $\Omega_{\text{plant}} \approx 1.05$.
6. **FFR $= T_{\text{v}}$.** Invasively measured FFR equals $1-\Gamma_{\text{v}}^2$ computed from CT angiography geometry.
7. **Cartilage as relay.** $Z_{\text{cart}} \approx \sqrt{Z_{\text{bone}} \cdot Z_{\text{syn}}}$.
8. **Vascular fractal dimension.** 3D micro-CT yields $d_H \to 3$ below capillary spacing.
9. **Organ hierarchy invariance.** MRI-based perfusion indices rank-order as Table IV predicts.
10. **Cross-kingdom β.** $\beta = n^{1/d}$ even in vascular plants.

XVII. CONCLUSION

Starting from a single blood vessel ($Z_{\text{v}} = 8\mu L/\pi r^4$) and the three constraints (C1–C3), this paper derives:

1. **Murray's Law** ($r_d = r_p/n^{1/3}$) as the exact minimum of the vascular action.
2. **Dual-network organ health** $H = (1-\Gamma_{\text{n}}^2)(1-\Gamma_{\text{v}}^2)$ from multiplicative transmission.
3. **Four failure modes** classified by which network fails.
4. **Fractal cost** $A_{\text{cut}} \propto M^{(d-1)/d}$—economies of scale from the relay hierarchy.
5. **Murray's ratio** $\beta = n^{1/d}$ as geometric inevitability.
6. **Kleiber's Law** $B \propto M^{3/4}$ from capillary invariance in $d = 3$.
7. **Dual-network packing overhead** $\Omega \approx 1.33$, with $\Omega_n \approx \beta = 2^{1/3}$.
8. **Interface universality**: one formula across seven physical domains and 14 orders of magnitude.

The title equation:

$$1_v \times 1_n = \Omega \approx \beta = 2^{1/3} \approx 1.26. \tag{27}$$

Life pays a fractal tax of $\beta - 1 \approx 26\%$ for the privilege of being simultaneously fed and informed.

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